Where the ECDLP exponent can move, and where it cannot

Status (assessment, 2026-10-18). Assessment from committed evidence; no new speedup is claimed. Every number below cites a frozen artefact in this repository. Fitted exponents are quoted with their intervals and the sizes they were fitted on; a `toy`-tier figure is never carried to another tier (`docs/bounds/README.md`). The companion sweep (`research/ecbench_comprehensive_sweep_20261018/`) refits the generic methods on prime-field and Koblitz curves at 2 targets × 3 rounds over nine and six sizes, and a fixed-base index calculus over six; its three sessions are sealed and replayed, and §5 reads them.

Cost is written `gae = C · r^α` in ecbench's counted unit (group-addition equivalents), `r` the prime subgroup order. "Moving the exponent" means `α < 1/2` on a domain, with the 95 % interval of the fit excluding `1/2`, over at least four sizes, every answer verified, and the whole process charged (`docs/bounds/README.md`, acceptance rule).

1. What the frontier says today

From `docs/bounds/FRONTIER.md` (frontier `ECFR1h8b45b672c283`, 46 bounds):

domain	method	α (95 %)	sizes ($\log_2 r$)	ops × floor (95 %)	bound
koblitz · medium	`rho.signed_frobenius_strong`	0.451 [0.399, 0.512]	32.1 … 47.2 (6)	1.089 [0.927, 1.261]	`ECBND1h05a2ea918f3c`
koblitz · medium	`bsgs.negation`	0.503 [0.474, 0.531]	32.1 … 47.2 (6)	7.669 [6.981, 8.375]	`ECBND1he836821517fb`
koblitz · medium	`claw.pair_table`	0.467 [0.423, 0.499]	32.1 … 47.2 (6)	2.390 [2.160, 2.639]	`ECBND1h25ac58612850`
koblitz · medium	`rho.negation`	0.473 [0.125, 0.553]	27.8, 32.1, 39.0	6.516 [5.780, 7.245]	`ECBND1h83091af19aee`
koblitz · medium	`kangaroo.vow`	0.486 [−0.312, 0.571]	27.8, 32.1, 39.0	15.558 [13.742, 17.597]	`ECBND1h7f1d4d59799f`
koblitz · toy	every generic method	intervals contain 0.5 and are wide	15–22	3.3 – 30	table 1 of the page
prime · toy	`ic.pipeline` prime-abscissa 32, MITM (negation-folded)	0.841 [0.500, 1.024]	17.7, 19.1, 21.2, 24.0	22.686 [11.421, 40.107]	`ECBND1hbab6d6dc37fc`
prime · toy	`ic.pipeline` prime-abscissa 32, subtract oracle	1.005 [0.786, 1.099]	17.7, 19.1, 21.2, 24.0	1258.2 [395.6, 2482.9]	`ECBND1h9d3beb6852c6`

Reading. Every generic method fits `α ≈ 1/2`, as it must (§2). The `claw.pair_table` interval `[0.423, 0.499]` sits just under `1/2` on six sizes; it is a √r-time, `r^{1/4}`-memory meet-in-the-middle whose fitted exponent drifts because its table share changes with `r`. Its ops are 2.4× the floor, so it is not a candidate: an exponent claim on it would be an artefact of the constant shifting across the range, the pattern `docs/bounds/README.md` warns of. The two prime-field index-calculus rows fit `α = 0.84 [0.50, 1.02]` and `1.01 [0.79, 1.10]` on four sizes at a fixed 32-point factor base: with the base held constant the decomposition step grows like `r/|F|^{m-1}`, so the fitted exponent is *above* `1/2`, the opposite direction from a win, and 23× and 1258× the floor in ops.

2. Why the generic exponent cannot move

For any algorithm that uses `E(F_p)` only through the group law and equality tests, Shoup (1997) gives `T ≥ Ω(√r)` online, and with `S` words of target-independent advice Corrigan-Gibbs–Kogan (2018) give `S·T² ≥ Ω̃(r)`. `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §1 prices this repository's own methods against that line: rho `S = 0`, `T = √(πr/4)`; preprocessing rho `S = r^{1/3}`, `T ≈ 2r^{1/3}` (tight); every index-calculus variant whose decomposition is a table look-up (`IC-2`, `IC-3`, `IC-m`) lands on or above `S·T² ≥ r/4`. So on prime-field curves every method in ecbench's registry is bound to `α = 1/2` on a single target, and the only movable quantities are:

- **the constant** through usable automorphisms: √2 from negation (`rho.negation`, `bsgs.negation`), √3 and √2 from the order-6 and order-4 CM automorphisms (`rho.cubic`, `rho.quartic`: measured 0.70 and 0.71 of `rho.negation`, `benchmarks/README.md`, 2026-10-17), `√(2n)` from signed Frobenius on Koblitz curves (`rho.signed_frobenius_strong` at 1.089× its own floor, medium tier);
- **the walk's distance to its floor**: the medium-tier Koblitz rho is within 9 % of `√(π r / 4n)`, the toy-tier generic rows 3–10× above theirs (start-up and distinguished-point overheads at `r < 2^22`);
- **accounting across targets**: Kuhn–Struik's `L(k)` for `k` targets (table C of the leaderboard) and Bernstein–Lange precomputation (`r^{1/3}` online after `r^{2/3}` setup) lower the per-target cost without lowering `S·T²`.

None of these is an exponent move, and the success criterion does not ask for one: each is a measured improvement over the standing baseline when it lowers end-to-end `S` on the same instances (AGENTS.md, "Success criterion").

3. Where the exponent is movable in principle, and what the evidence

says

3.1 Koblitz curves over F_{2^n}: Frobenius-invariant index calculus

The lever is real (RESEARCH_KOBLITZ_INDEX_CALCULUS.md, "Where the speed-up lives"): a Frobenius-invariant factor base collapses |F| unknowns to |F|/n orbit unknowns, saving n relations and n² in the matrix, and m! in the decomposition. The measured record (docs/ic/BOUNDARY_TARGETS_CHARGED_SUPPLEMENT_20261008.md, vs_rho rows):

round	best cold row	fitted total α (R^2, sizes)	× counted signed-Frobenius rho
1 (2026-09-21)	n = 41 signed-orbit fold, S = 58.8	0.54 (0.71, 10 rungs); 0.51 (0.98, cofactor ≤ 4)	300×
2	folded pair table, walked targets, S = 5.12	0.43 (0.82, 4 rungs)	26.2×
3–5	restart at one addition, lazy offsets, S = 5.11	unchanged family	26.1×
leaderboard table A (rule v3)	n = 47, S = 9.51 vs reference 1.82	–	5.23×
§20, k = 32 batch (research/ic_exponent_20260926)	thread / batch rho 4.86× [4.21, 5.55] at 2^39	ratio exponent β = 0.137 [−0.10, 0.37]	13.6× at 2^47.2

Reading. The best fitted total exponent, 0.43 over four rungs with $R^2 = 0.82$, has no interval narrow enough to exclude 1/2, and the family's derived optimum is itself ≈ 30× the asymptotic signed-Frobenius rho (round 3). The §20 ratio exponent β contains 0, so "IC/rho shrinks with r" is not established either. What stops the exponent from moving is named precisely in the Koblitz note: every oracle works at m = 2, 3, a useful attack needs m ≈ n/ℓ, and the chaining representation needs unknowns(n, ℓ, m) = m·ℓ + (m − 2)·n variables (633 at n = 63, ℓ = 6 against a 64-variable budget), while the measured first fall degree never exceeds 3. The algebra is benign; the variable count is the wall. **This is the one place in the program where the exponent is movable with known mathematics (Õ(2^{(2−2/m)·n/…}) for m summands) and where the blocker is a representation problem, not a lower bound.**

3.2 Extension-field curves: Gaudry's k = 4 and the F_(2^m)^4

analogue ic.k4_gaudry and ic.pipeline on fp4_translation and F2m4 workloads implement Gaudry–Diem index calculus with m = n = 4 summands. Its heuristic cost Õ(p^{2−2/n}) = Õ(p^{1.5}) against rho's √r ≈ p^2 is α = 3/8 in r: an exponent move on paper, at constants (the S_5 summation polynomial and its Gröbner/root-finding step) that this repository has measured only on toy fields. These are the correct domains for an exponent claim because the sub-1/2 exponent is a theorem-level heuristic, not a hope; the work is to drive the constant down and run four sizes with the whole process charged. The research/ecbench_rootfinding_* studies (2026-10-12…14) and research/gaudry_quartic_d4_ecbench_20261013 are the standing baselines.

3.3 Prime-field curves: closed on paper, probes negative

RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md §4 lists the only levers with a falsifier each: small roots beyond Coppersmith for the S_{m+1} shape (milestone δ ≥ 1/4 at m = 3, target δ ≈ 1/5 at m = 4; Probes A–C negative), a factor base with sub-B-degree membership (none exists in F_p), transfer to a structured setting (the isogeny volcano studies of 2026-10-16/17 find no level effect: floor/crater 1.022 [0.983, 1.062]), and the closed-on-paper items of §4.6 (symmetric S_3, subgroup bases, xedni, embedding into an extension, which costs p^{2−2/n} against √p and enlarges the problem). The leaderboard's prime-field IC rows (table B) sit 5.5–13.5× rho with their family optimum O(r^{1/6}) above a flat reference: no crossover at any size. **No exponent move is available here; the honest program is constants (§2) and accounting.**

4. Recommendations, ranked by expected movement per unit of work

1. **Koblitz chaining at m ≥ 4 with a variable budget that scales.** The wall is unknowns = m·ℓ + (m − 2)·n. Replace the full chain with the Frobenius-orbit-reduced chain (unknowns per orbit, not per point) and measure the first fall degree and solve time at m = 4, 5 on n = 23, 29, 31, 37; a four-size fit with the interval of α excluding 1/2 is the acceptance test. Protocol: ecbench challenge against ECBND1h05a2ea918f3c (the medium-tier rho holder).
2. **Gaudry k = 4 end to end at four field sizes.** ic.k4_gaudry on fp4_translation with the root-finding oracle from the 2026-10-13 tuning; whole process charged; aim for the α = 3/8 fit to exclude 1/2. Even at constants far above rho this is the first row the program can label *exponent-level* honestly.
3. **Generic constants to the floor.** The toy-tier generic rows sit 3–10× above √(π r / 2A); the medium-tier Koblitz rho is at 1.09×. The companion sweep quantifies the prime-field gap at 2^16–2^32 (§5). Start-up (two scalar multiplications per walk), distinguished point density and the jump-table build are the whole of that gap; none is an exponent, all are improvements under the success criterion.
4. **Kernel engineering with identical counts** (the two companion passes of this date on the Rust and C kernels): wall-clock only, no change to any S, recorded as engineering.

5. Companion sweep (filled from the sealed sessions)

`research/ecbench_comprehensive_sweep_20261018/README.md` holds three sealed sessions of this date, every run verified and every measured run replayed identically (`ECBS1ha345d9b7f6b2`, `ECBS1h88c58e3e8c89`, `ECBS1hec2206c6baf4`; 702 measured runs, 702 replays). Descriptive OLS exponents, `α ± se`:

domain	method	α ± se (sizes)	mean S range across sizes
prime, `2^16 … 2^32`	`rho.plain`	0.474 ± 0.013 (9)	2.07 ⟶ 1.35 (floor 0.886)
prime	`rho.negation`	0.447 ± 0.012 (9)	1.91 ⟶ 0.91
prime	`bsgs.textbook`	0.499 ± 0.014 (9)	1.14 … 1.82
prime	`bsgs.interleaved`	0.501 ± 0.023 (9)	0.85 … 1.84
prime	`bsgs.negation`	0.498 ± 0.022 (9)	0.64 … 1.32
prime	`kangaroo.vow`	0.487 ± 0.032 (9)	1.07 … 3.00
Koblitz `K_1`, `2^15 … 2^37`	`rho.signed_frobenius_strong`	0.248 ± 0.015 (6)	4.54 ⟶ 0.11 (floor 0.215 ⟶ 0.129)
Koblitz	`rho.negation`	0.430 ± 0.029 (6)	1.97 ⟶ 1.08
Koblitz	`bsgs.negation`	0.511 ± 0.015 (6)	0.84 … 1.35
Koblitz	`kangaroo.vow`	0.562 ± 0.021 (6)	0.79 … 2.97
prime, `2^16 … 2^25`	`ic.pipeline`, prime-abscissa 32, MITM	0.757 ± 0.051 (6)	10.9 ⟶ 59.0; 5.7× ⟶ 43× `rho.negation`
prime	`rho.frozen_reference`	0.404 ± 0.037 (6)	5.24 ⟶ 2.17

Reading. The BSGS and kangaroo fits contain `1/2`. The two rho fits below `1/2` are the start-up artefact of §1: `S` falls across the range while the floor is constant, because the two scalar multiplications per walk and the jump-table build are a larger share at `2^16` than at `2^32`. The Koblitz strong reference shows it at scale (32 lanes of start-up before a step; 20× the floor at `2^16`, at the floor by `2^32`), so its `0.248` is not an exponent and the medium-tier frontier fit `0.451 [0.399, 0.512]` on `2^32 … 2^47` stands as the admitted value. The fixed-base index-calculus row fits above `1/2`, consistent with the frontier's `0.841 [0.500, 1.024]` on the same configuration, and its gap to rho widens with `r`. Nothing in the sweep moves an exponent; it quantifies the generic constants recommendation 3 names: at `2^28 … 2^32` the tuned rho arms are within their own model's expectation (1.25 and 0.89) and the gap to the floor is 1.0 − 1.5× in the counted unit.

6. Graphs

`exponent_map.svg` (beside this file, from `exponent_map.py`, with the plotted values in `exponent_map.diagram.json`): the exponent α per domain and method with its interval against the `1/2` line, by evidence class (frontier bound with its bootstrap interval, this sweep with `± 1.96 se`, fitted without a published interval, on paper). `REPORT.pdf` embeds it. The sweep's own graphs (mean `S` per arm and size, distributions, phases, scaling fits) are under `research/ecbench_comprehensive_sweep_20261018/reports/*/`.

References inside this repository

- `docs/bounds/FRONTIER.md`, `docs/bounds/README.md`
- `docs/ic/LEADERBOARD.md`, `docs/ic/BOUNDARY_TARGETS_CHARGED_SUPPLEMENT_20261008.md`
- `research/ic_exponent_20260926/README.md`, `analysis.json`
- `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md`, `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`
- `benchmarks/README.md` rows of 2026-10-16/17 (volcano studies, `rho.cubic`/`rho.quartic`)

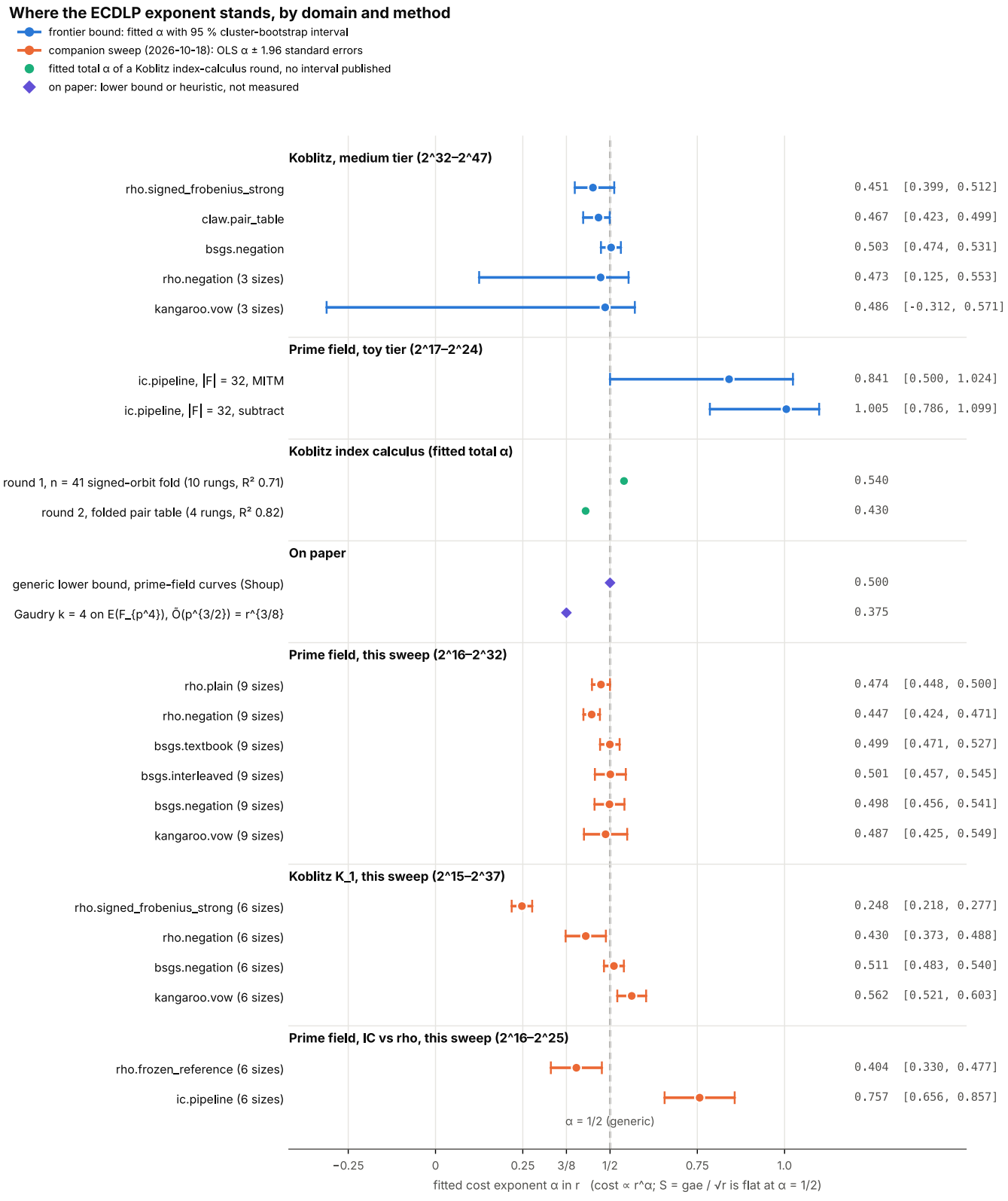


Figure 1: The ECDLP cost exponent α by domain and method, with intervals, against α = 1/2. Frontier bounds carry their 95 % cluster-bootstrap interval; the companion sweep's fits carry ± 1.96 standard errors; the Koblitz index-calculus rounds and the on-paper values have none. Values in `exponent_map.diagram.json`.